

df , Δf
 $\underline{\underline{df}}$ $\rightarrow$ infinitesimal
 $\textcircled{dt}$
 $\uparrow$
 Δf $\rightarrow$ practical
 $\textcircled{\Delta t}$

$$V = f(S, t)$$

$S \rightarrow S + \Delta S \rightarrow \1
 $t \rightarrow t + \Delta t$
 $\searrow$ $\underline{\underline{1 \text{ day}}}$

$$\Delta V =$$

$$\frac{\partial^2 f}{\partial W^2} \cdot \textcircled{\underline{\underline{dw}}}^2 \times \textcircled{dt}$$

$$\textcircled{dV}$$

$$\textcircled{dt}$$

w_t

Case I. $F = f(w)$

(a) $F = w^2$

$$dF = \frac{dF}{dw} (dw) + \frac{1}{2} \frac{d^2 F}{dw^2} (dw)^2$$

$$= 2w \cdot dw + \frac{1}{2} \times 2 \times dt$$

$$= 2w dw + dt \quad (\text{not a martingale})$$

(b) $f = e^w$, $\frac{dF}{dw} = e^w$, $\frac{d^2 F}{dw^2} = e^w$

$$dF = \frac{dF}{dw} (dw) + \frac{1}{2} \frac{d^2 F}{dw^2} (dw)^2$$

$$= e^w \cdot dw + \frac{1}{2} e^w \cdot dt$$

$$= e^w (dw + \frac{1}{2} dt) = F (dw + \frac{1}{2} dt)$$

$$dF/F = dw + \frac{1}{2} dt$$

$$\textcircled{2} \quad F = \underline{w^2 - t}, \quad \frac{\partial F}{\partial t} = -1, \quad \frac{\partial^2 F}{\partial t^2} = 0$$

$$\frac{\partial F}{\partial w} = 2w, \quad \frac{\partial^2 F}{\partial w^2} = 2$$

$$dF = \frac{\partial F}{\partial t}(dt) + \frac{\partial F}{\partial w}(dw) + \frac{1}{2} \frac{\partial^2 F}{\partial t^2} (dt)^2 + \frac{1}{2} \frac{\partial^2 F}{\partial w^2} (dw)^2 + \frac{\partial^2 F}{\partial t \partial w} (dt dw)$$

$$= (-1)dt + 2w dw + \frac{1}{2} \times 2 \times dt$$

$$= \underline{2w dw} \quad (\text{Martingale!!})$$

$$\int_0^+ w dw = \frac{1}{2} (w_t^2 - t) \rightarrow$$

$$\int_0^+ dF = 2 \int_0^+ w dw$$

$$\Rightarrow F(t) - F(0) = 2 \int_0^+ w dw$$

$$\Rightarrow (w_t^2 - t) - (\underline{w_0^2} - 0) = 2 \int_0^+ w dw$$

$$\Rightarrow \int_0^+ w dw = \frac{1}{2} (w_t^2 - t)$$

$$(b) \quad F = te^w, \quad \frac{\partial F}{\partial t} = e^w, \quad \frac{\partial F}{\partial w} = te^w, \quad \frac{\partial^2 F}{\partial w^2} = te^w$$

$$dF = e^w \cdot dt + te^w \cdot dw + \frac{1}{2} te^w \cdot dt$$

(ln S)

$$ds_t = \mu(s_t)dt + \sigma(s_t)dW_t$$

$$\underline{\underline{ds_s = \mu dt + \sigma dw.}}$$

$$\underline{F = F(S) = \log S} \rightarrow$$

$$S_t = ??$$

$$dF = \frac{dF}{ds}(ds) + \frac{1}{2} \frac{d^2 F}{ds^2}(ds)^2$$

$$= \frac{1}{s} \cdot ds - \frac{1}{2s^2} (ds)^2$$

$$\frac{dF}{ds} = \frac{1}{s}, \quad \frac{d^2 F}{ds^2} = -\frac{1}{s^2}$$

$$= \mu dt + \sigma dw - \frac{1}{2}(\sigma^2 dt) = (\mu - \frac{1}{2}\sigma^2)dt + \sigma dw$$

$$\int_0^t d(\log s) = \int_0^t (\mu - \frac{1}{2}\sigma^2) dt + \int_0^t \sigma dw$$

$$\Rightarrow \log(S_t) - \log(S_0) = (\mu - \frac{1}{2}\sigma^2)t + \sigma W_t$$

$$\Rightarrow \log(S_t/S_0) = (\mu - \frac{1}{2}\sigma^2)t + \sigma W_t$$

$$\rightarrow \underline{\underline{S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}}}, \quad W_t \sim N(0, t)$$

$$\rightarrow \log(S_t) = \log(S_0) + (\mu - \frac{1}{2}\sigma^2)t + \sigma W_t$$

$$\underline{\underline{S_t = (S_0) e^{X_t}}}, \quad X_t = (\mu - \frac{1}{2}\sigma^2)t + \sigma W_t$$

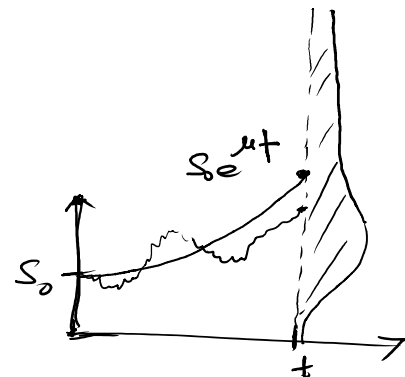
$$\rightarrow X_t \sim N((\mu - \frac{1}{2}\sigma^2)t, \sigma^2 t)$$

$$S_t \sim \text{lognormal}((\mu - \frac{1}{2}\sigma^2)t, \sigma^2 t), \quad \underline{\underline{GBM}}$$

$$\mathbb{E}[\log(S_t)] = \log(S_0) + (\mu - \frac{1}{2}\sigma^2)t$$

$$\mathbb{E}[S_t] = (S_0) e^{(\mu - \frac{1}{2}\sigma^2)t + \frac{1}{2}\sigma^2 t} = S_0 e^{\mu t}$$

$$\underline{\underline{Y_t = S_0 e^{-\frac{1}{2}\sigma^2 t + \sigma W_t}}}, \quad \begin{matrix} \text{exponential} \\ \text{(Martingale)} \end{matrix}$$



$$Y_t = \underline{S_0} e^{\underline{-\frac{1}{2}\sigma^2 t + \sigma W_t}} \quad (\text{Martingale})$$

$$\mathbb{E}[Y_t] = S_0 \cdot e^{-\frac{1}{2}\sigma^2 t + \frac{1}{2}\sigma^2 t} = S_0 \cdot e^0 = \underline{S_0}$$

$$dw_1, dw_2, \quad dw_1 \sim N(0, dt) \\ dw_2 \sim N(0, dt)$$

$$\mathbb{E}[dw_1 \cdot dw_2] = \rho dt$$

$$\mathbb{V}[dw_1 \cdot dw_2] \rightarrow 0$$

$$\langle dw_1 \cdot dw_2 = \rho dt \rangle$$

$$\checkmark \text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] \rightarrow$$

$$\checkmark \mathbb{V}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

$$\text{Cov}(dw_1, dw_2) = \mathbb{E}[dw_1 \cdot dw_2] - \mathbb{E}[dw_1] \cdot \mathbb{E}[dw_2]$$

$$\rho \times \sqrt{dt} \cdot \sqrt{dt} = \mathbb{E}[dw_1 \cdot dw_2] = \rho dt$$

$$F(s_1, s_2) \rightarrow ds_1 = s_1 \mu_1 dt + s_1 \sigma_1 dw_1, \quad dw_1 \cdot dw_2 = \rho dt$$

(basket products)

$$\rightarrow ds_2 = s_2 \mu_2 dt + s_2 \sigma_2 dw_2$$

$$dF = \frac{\partial F}{\partial s_1} ds_1 + \frac{\partial F}{\partial s_2} ds_2 + \frac{1}{2} \frac{\partial^2 F}{\partial s_1^2} (ds_1)^2 + \frac{1}{2} \frac{\partial^2 F}{\partial s_2^2} (ds_2)^2 + \frac{\partial^2 F}{\partial s_1 \partial s_2} (ds_1)(ds_2)$$

$$= \left(\frac{\partial F}{\partial s_1} \right) ds_1 + \frac{\partial F}{\partial s_2} ds_2 + \frac{1}{2} \frac{\partial^2 F}{\partial s_1^2} s_1^2 \sigma_1^2 dt + \frac{1}{2} \frac{\partial^2 F}{\partial s_2^2} s_2^2 \sigma_2^2 dt + \frac{\partial^2 F}{\partial s_1 \partial s_2} s_1 s_2 \sigma_1 \sigma_2 \rho dt$$

$$\begin{aligned}
 F &= xy \rightarrow \\
 dF &= \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy + \frac{1}{2} \frac{\partial^2 F}{\partial x^2} (dx)^2 + \frac{1}{2} \frac{\partial^2 F}{\partial y^2} (dy)^2 + \frac{\partial^2 F}{\partial x \partial y} dx dy \\
 &= y dx + x dy + 0 + 0 + dx dy \\
 &= \underline{y dx + x dy + \underline{dx \cdot dy}}
 \end{aligned}$$

$$F = \frac{x}{y} = x \cdot \left(\frac{1}{y}\right), \quad \frac{\partial F}{\partial x} = \frac{1}{y}, \quad \frac{\partial F}{\partial y} = -\frac{x}{y^2}, \quad \frac{\partial^2 F}{\partial x^2} = 0, \quad \frac{\partial^2 F}{\partial y^2} = \frac{2x}{y^3}, \quad \frac{\partial^2 F}{\partial x \partial y} = -\frac{1}{y^2}$$

$$\begin{aligned}
 dF &= \frac{1}{y} dx - \frac{x}{y^2} dy + \frac{1}{2} \cdot \frac{2x}{y^3} (dy)^2 - \frac{1}{y^2} dx \cdot dy \\
 &= \left(\frac{x}{y}\right) \left(\frac{dx}{x} - \frac{dy}{y} + \left(\frac{dy}{y}\right)^2 - \left(\frac{dx}{x}\right)\left(\frac{dy}{y}\right)\right) \rightarrow d\left(\underline{\underline{s_t / B_t}}\right)
 \end{aligned}$$

$$F = F(w)$$

$$\int_0^t dF = \int_0^t \left(\frac{dF}{dw} \right) dw + \int_0^t \frac{1}{2} \left(\frac{d^2 F}{dw^2} \right) \cdot dt$$

$$\Rightarrow F(t) - F(0) = \int_0^t \left(\frac{dF}{dw_s} \right) dw_s + \frac{1}{2} \int_0^t \frac{d^2 F}{dw_s^2} ds$$

$$\int_0^t \left(\frac{dF}{dw_s} \right) dw_s = F(t) - F(0) - \frac{1}{2} \int_0^t \frac{d^2 F}{dw_s^2} ds$$

$$\int_0^t e^w dw = ?$$

$$\frac{dF}{dw} = e^w \Rightarrow F = e^w$$

$$= e^{w_t} - e^{w_0} - \frac{1}{2} \int_0^t e^{w_s} ds$$

$$= e^{w_t} - 1 - \frac{1}{2} \int_0^t e^{w_s} ds$$

$$\underline{\underline{w_0 = 0}}$$

$$F(t, w).$$

$$\int dF = \int \frac{\partial F}{\partial t} dt + \int \frac{\partial F}{\partial w} dw + \int \frac{1}{2} \frac{\partial^2 F}{\partial w^2} (dt)$$

$$\underline{\underline{F(t) - F(0)}} = \int_0^t \left(\frac{\partial F}{\partial s} + \frac{1}{2} \frac{\partial^2 F}{\partial w_s^2} \right) ds + \int_0^t \underbrace{\left(\frac{\partial F}{\partial w_s} \right)}_{\text{Question}} dw_s$$

$$+ \int_0^t \underline{\underline{(sw + e^w)}} dw$$

$$\frac{\partial F}{\partial w} = sw + e^w$$

$$\Rightarrow F = \underline{\underline{sw^2/2 + e^w}}$$

$$dx_t = -rx_t dt + \sigma dw_t, \quad x_t = ?, \quad S_t = ?$$

$$\Rightarrow dx_t + \underline{r}x_t dt = \sigma dw_t$$

$$\left(\frac{dx_t}{dt} + \underline{r}x_t = \sigma \frac{dw_t}{dt} \right)$$

$$\frac{dy}{dx} + P(x)y = Q(x)$$

$$\text{IF} = e^{\int P dx}$$

$$\int r dt = e^{rt}$$

$$e^{rt} \cdot dx_t + r e^{rt} \cdot x_t dt = \sigma e^{rt} \cdot dw_t$$

$$\Rightarrow \int_0^t d(e^{rs} \cdot x_s) = \int_0^t \sigma e^{rs} \cdot dw_s$$

$$\Rightarrow [e^{rt} \cdot x_t]_0^t = \sigma \int_0^t e^{rs} dw_s$$

$$\Rightarrow \underline{e^{rt} \cdot x_t} - x_0 = \sigma \int_0^t e^{rs} dw_s$$

$$\Rightarrow \underline{x_t} = \underline{x_0 e^{-rt}} + \underbrace{\sigma \int_0^t e^{-r(t-s)} dw_s}_{Y. \text{ (Ito integral)}}$$

$$Y \sim N\left(0, \int_0^t Y^2 ds\right)$$

$$\int_0^t e^{-2r(t-s)} ds$$

$$\begin{aligned} t-s &= u \\ ds &= -du \end{aligned}$$

$$= \int_0^t e^{-2ru} du = \left[\frac{e^{-2ru}}{-2r} \right]_0^t = \frac{1}{2r} (1 - e^{-2rt})$$

$$x_t \sim N\left(\underline{x_0 e^{-rt}}, \frac{\sigma^2}{2r} (1 - e^{-2rt})\right)$$

$$t \rightarrow \infty, \quad x_\infty \sim N\left(\underline{0}, \frac{\sigma^2}{2r}\right)$$

